AE 481 Team 01 Assignment 10

November 23, 2025

1 V-n Diagrams

V-n diagrams were created for the F/A-67 at three flight conditions:

- Maximum mission weight (air-to-air)
- Mid-mission weight (air-to-air)
- Minimum mission weight (strike)

The mission profile with the heaviest takeoff weight is air-to-air, and the mission profile with the lower minimum weight is strike. Mid-mission weight was determined for the air-to-air mission due to the increased maneuverability requirements for air-to-air combat. Mid-mission weight is defined when the aircraft enters combat. The maximum positive limit load factor N_z was set using RFP requirements, and feasibility from sizing plot analyses. Aerodynamically, the aircraft can meet the desired 8g maximum positive load factor. For maximum negative limit load factor, a value of -6g was selected based on recommendations from Raymer [1].

The calculation of maneuver loads and gust loads were determined using equations from Raymer [1]. The load factor n for maximum lift was calculated as

$$n = \frac{1}{2W} \rho_{\text{sl}} V_{\text{EAS}}^2 S C_{L,\max} \tag{1}$$

where sea level density and equivalent air speed are used. The corner speed V_A was calculated using equation (2). The positive and negative limit load factors use an identical $C_{L,\max}$ value. This is assumed since the F/A-67 airfoil is nearly symmetric.

$$V_A = \sqrt{\frac{2WN_z}{\rho_{\text{sl}} S C_{L,\max}}} \tag{2}$$

The maximum cruise speed V_C was selected as the design air-to-air supersonic dash speed of Mach 1.65. The maximum dive speed V_D was selected as the desired RFP dash speed of Mach 2.0. Velocities were determined at the dash altitude of 30,000 ft and were converted to equivalent airspeeds using equation (3).

$$V_{\text{EAS}} = \sqrt{\frac{\rho}{\rho_{\text{sl}}}} V \tag{3}$$

For the V-n diagram, the negative maneuver limit load factor between V_C and V_D increases from the maximum negative load factor to a value of -1g. This was determined based on guidelines for fighter aircraft in MIL-A-8861B [2]. A safety factor of 1.5 is applied to the maximum limit load factor to give the ultimate load factor.

1.1 Gust Loads

Gust loads were calculated for rough air, at V_C, and at V_D, at 20,000 ft. The gust load factor is calculated using equations (4)-(7) [1]. The load factor is defined as

$$n = 1 \pm \frac{\rho K U_{\mathrm{de}} V C_{L_\alpha}}{2W/S} \tag{4}$$

The parameter K is determined based on if the aircraft is subsonic or supersonic. For a subsonic gust,

$$K = \frac{0.88\mu}{5.3 + \mu} \tag{5}$$

For a supersonic gust,

$$K = \frac{\mu^{1.03}}{6.95 + \mu^{1.03}} \tag{6}$$

The mass ratio μ is defined as

$$\mu = \frac{2W/S}{\rho g \bar{c} C_{L_\alpha}} \tag{7}$$

The C_{L_α} of the aircraft was found to be 5.028 using AVL [3]. The gust vertical velocity, U_{de}, was determined using Raymer [1] and are listed in Table 1. The rough air speed V_B was determined as the intersection of the positive gust line and the maximum lift maneuver load line.

Table 1: Gust vertical velocities for different flight conditions.

	Gust velocity (ft/s)
Rough air speed (V_B)	66
Cruise speed (V_C)	50
Dive speed (V_D)	25

1.2 Results

The V-n diagrams for maximum weight (Fig. 1), mid-mission weight (Fig. 2), and minimum weight (Fig. 3), indicate all relevant speeds. Gust loads are plotted as dash-dotted lines. The ultimate maneuver load factor is plotted as a dotted line. The solid black lines indicate the combined flight envelope of limit maneuver loads and gust loads. The aircraft meets the desired design vertical load factor of 8g. Aircraft empty weight estimation also takes into account a 1.5 safety factor on the design load factor, which indicates the aircraft sizing accounts for the structural weight required to withstand combined loads envelope.

Since the combined load envelope does not exceed the ultimate load factor (including gusts), and the aircraft weight is sized using the ultimate load factor, the team is confident that the F/A-67 structure can withstand the design maneuver loads for all flight conditions.

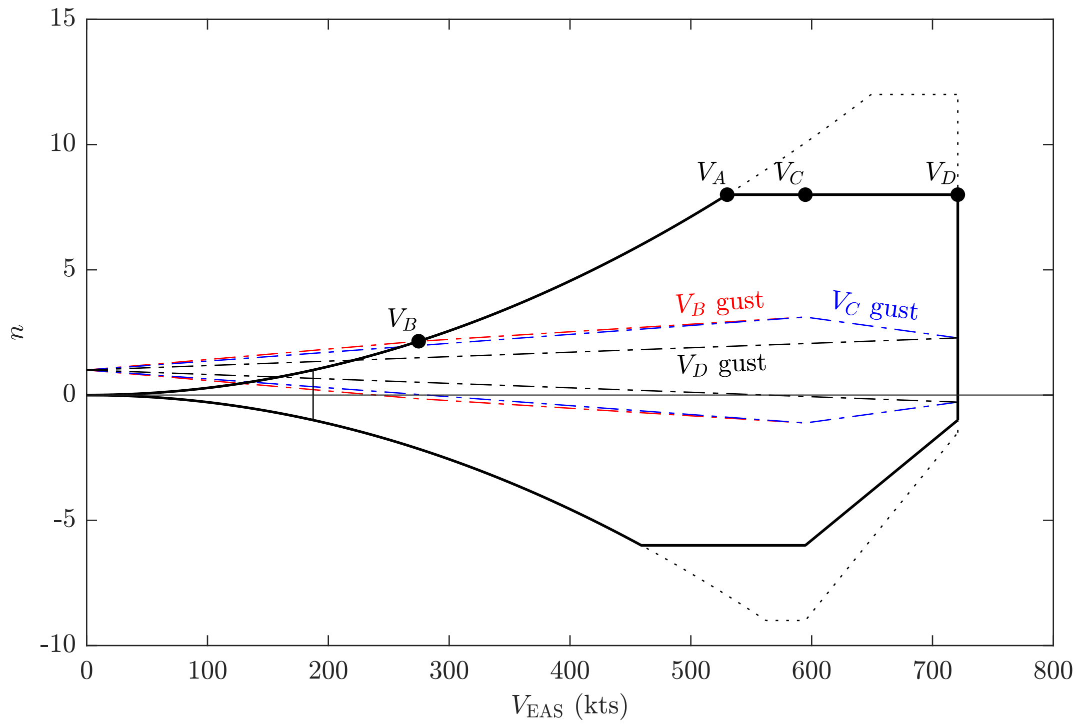

Figure 1: The V-n diagram at maximum weight. The solid black lines are the combined loads, the dotted line is the ultimate maneuver load, and the dash-dotted lines are gust loads. The 1g stall speed is denoted at the thin vertical line.

2 $C_{L,\max}$ Estimation

Because of the wing airfoil's sharp leading edge and high Reynolds number, it cannot be analyzed by XFOIL or ADflow at high angles of attack. Instead, experimental data can be used to approximate the $C_{\ell_{\max}}$. In NACA TN2502 [4], the stall behavior of the NACA 64A006 airfoil is studied at a Reynolds number of 5.8×10^6. This is assumed to behave similarly to the wing airfoil, as it exhibits leading edge stall. The studied Reynolds number is also comparable to the Reynolds number at takeoff of 17×10^6. The $C_{\ell_{\max}}$ of the NACA 64A006 is 0.8865.

The clean $C_{L_{\max}}$ of the aircraft is estimated using AVL [3] using the critical section method. The C_L is increased until the wing C_ℓ exeeds 0.8865. This also includes tail trim. The clean $C_{L_{\max}}$ is found to be approximately 0.79.

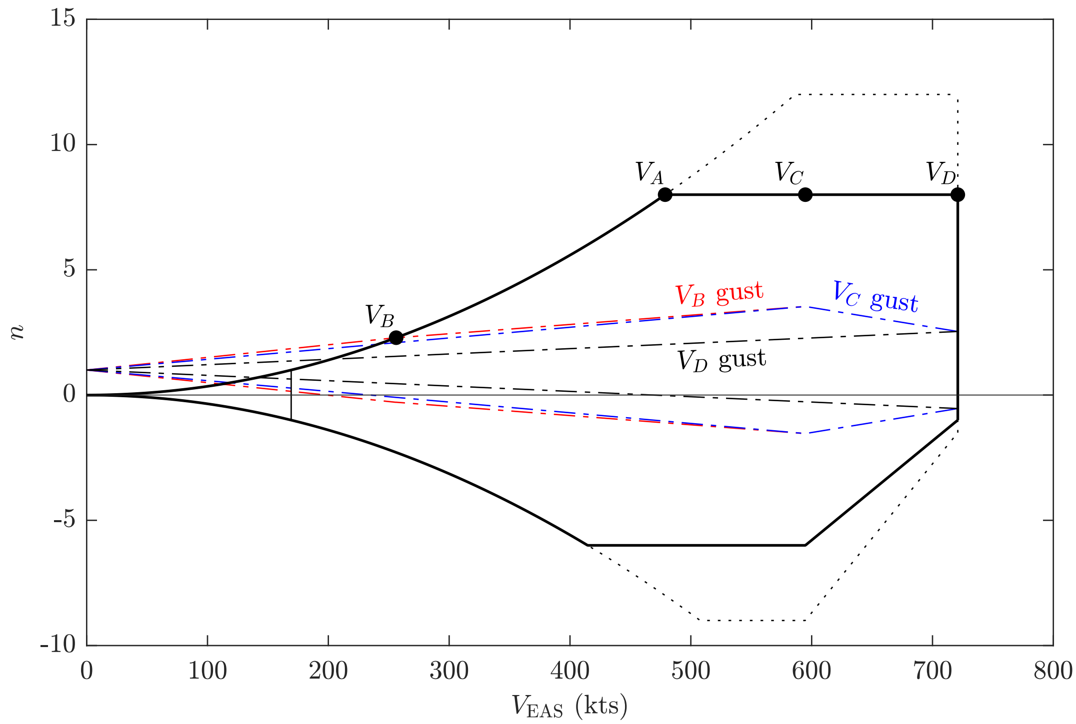

Figure 2: The V-n diagram at mid-mission weight. The solid black lines are the combined loads, the dotted line is the ultimate maneuver load, and the dash-dotted lines are gust loads. The 1g stall speed is denoted at the thin vertical line.

3 Group Work Split

Table 2: Work Split

Person	Work Percentage	Brief Description of Work Done
Michael Chen	16.67%	V-n diagram code and report writing.
Charles Choi	16.67%	Landing gear sizing and aircraft fuel tank sizing.
Cristina Erskine	16.67%	Helped find information on V-n diagrams, outlined CDR work.
James Gold	16.67%	Found diameter of landing gear, helped find information on V-n diagrams, assisted with report writing.
Santiago Ramos-Assam	16.67%	Refined wing CAD and modeled external stores carriage.
Thomas Sheridan	16.65%	Determined aircraft maximum lift coefficient and wrote report.

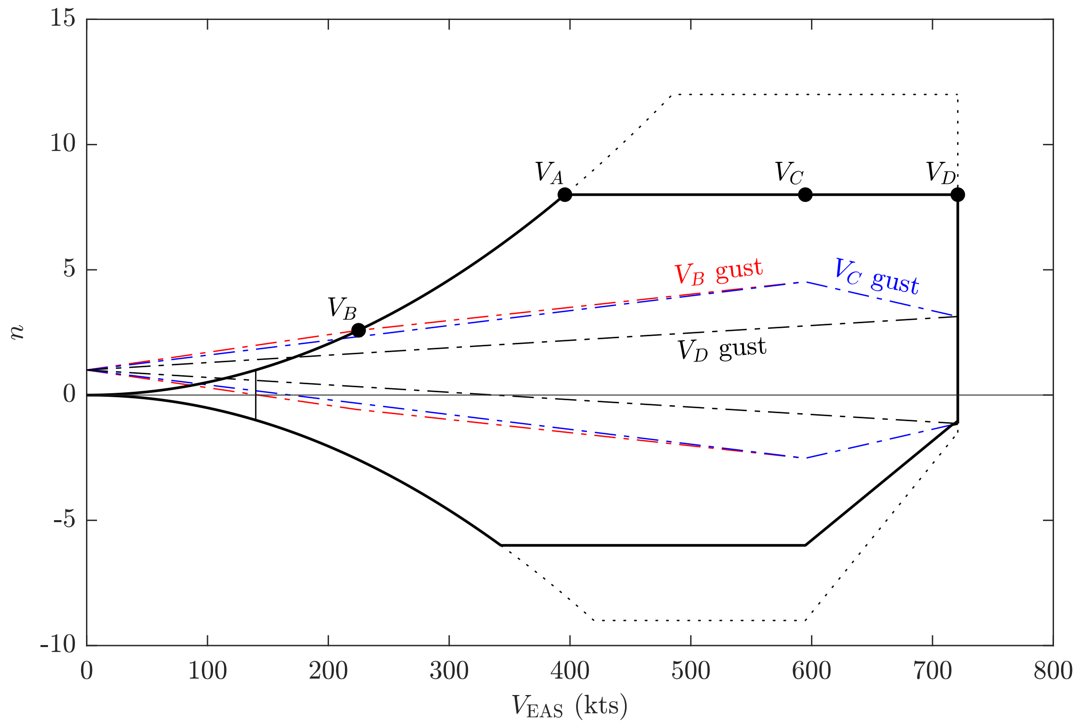

Figure 3: The V-n diagram at minimum weight. The solid black lines are the combined loads, the dotted line is the ultimate maneuver load, and the dash-dotted lines are gust loads. The 1g stall speed is denoted at the thin vertical line.

References

[1] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, 2018. ISBN: 9781624104909.

[2] U.S. Department of Defense. *MIL-A-8861B: Aircraft Strength and Rigidity Flight Loads*. 1986.

[3] *AVL 3.40*. 2022. URL: `https://web.mit.edu/drela/Public/web/avl/avl_doc.txt`.

[4] George B. McCullough and Donald E. Gault. *EXAMPLES OF THREE REPRESENTATIVE TYPES OF AIRFOIL-SECTION STALL AT LOW SPEED*. Tech. rep. NACA, 1951.